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RESEARCH NOTE

The Lower Cretaceous reference pole for North China, and its tectonic implications

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SUMMARY

A palaeomagnetic study of Lower Cretaceous continental sediments from the south–southwestern part of the Ordos basin, North China block (NCB), has been carried out. Stepwise thermal demagnetization was used to isolate magnetic components. The low-temperature component is generally removed between 200 °C and 350 °C and corresponds to the Earth's present field direction. A high-temperature component of magnetization, with two polarities, was defined from 10 sites with D = 16.4°, I = 50.5° ($K = 50.8$, $\alpha_{95} = 6.8°$). The corresponding pole position is located at 75.8°N, 208.7°E ($d_p = 6.1°$, $d_m = 9.1°$). This Lower Cretaceous pole compares well with the hairpin loop of the Eurasian synthetic APW path (Besse & Courtillot 1991) that is inferred from the North American, Eurasian, African and Indian plates. The available late Mesozoic palaeomagnetic data from the NCB and adjacent South China block (SCB) are reviewed. Comparison of these data suggests important intraplate deformation: the area to the east of the Tancheng–Lujiang fault may have undergone local clockwise post-Cretaceous rotations of 10–29° relative to the stable western craton of the NCB. The data also indicate that no palaeomagnetically significant eastward movement of the SCB with respect to the NCB and Eurasia occurred after the Cretaceous.

Key words: China, Lower Cretaceous, palaeomagnetic reference pole, Tancheng–Lujiang fault.

INTRODUCTION

Eastern Asia is considered to be an assemblage of continental fragments accreted to the Eurasian continent in the Mesozoic, as shown by the results of geological and palaeomagnetic studies (e.g. McElhinny *et al.* 1981; Zhang, Liou & Coleman 1984; Hsü *et al.* 1987; Ren *et al.* 1990; Zhao *et al.* 1990a,b; Enkin *et al.* 1992; Yang *et al.* 1992). The continental blocks present distinctive tectonic or stratigraphic characteristics and are usually separated by important deformation zones, including fold belts and fault zones, some with ophiolites or other indicators of sutures (Fig. 1a). The Qinling fold belt, for example, marks the boundary between the North China block (NCB) and the South China block (SCB). It is considered to be a collision-type orogenic belt, and is divided into several tectonic units. It includes the Palaeozoic continental margins of both the NCB and SCB and comprises a metamorphic complex with ophiolitic blocks and flysch nappes (e.g. Mattauer *et al.* 1985; Hsü *et al.* 1987). The fold belts developed during thrusting and magmatic activity associated with polyphase tectonic events, and many of these zones were reactivated at the time of the collision and penetration of the Indian subcontinent during the Tertiary (Tapponnier *et al.* 1982; Ren *et al.* 1990). Several faults such as the Red River fault that separate the SCB from the Indochina block, and the Altyn Tagh–North Qinling faults that intersect the fold belt between the NCB and the SCB, became strike–slip faults. As an example, these two zones underwent displacements of up to 600 km during the Tertiary. Geological field observations also reveal strike–slip faulting activity, which results in block rotations (e.g. Peltzer *et al.*

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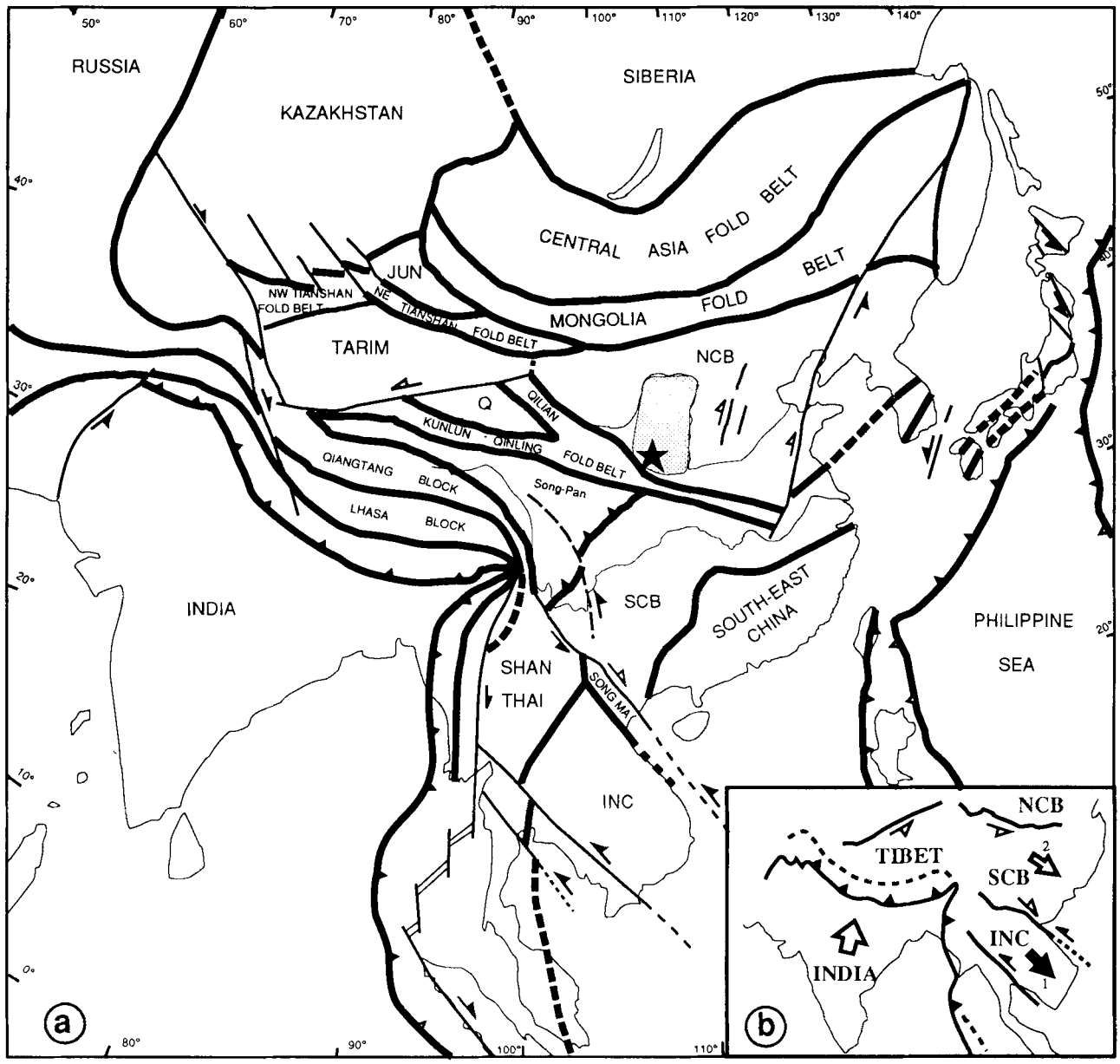


Figure 1. (a) Tectonic subdivisions of the main continental blocks and fold belts of eastern Eurasia: (b) simplified extrusion model of Asia; white arrows qualitatively represent major block motions with respect to Siberia. Arrows correspond to the senses and extrusion phases of the strike–slip motion on the major faults. Numbers refer to extrusion phase: 1, 50–20 Ma; 2, 20 Ma to 0 BP (after Tapponnier *et al.* 1982). The sampling area in the Ordos basin is indicated by a star symbol.

1985; Armijo, Tapponnier & Han 1989; Leloup 1991) (Fig. 1b). The palaeomagnetic method provides a quantitative tool to test the tectonic interpretation of displacements and rotations between eastern Eurasian blocks. A number of published palaeomagnetic studies have addressed the collision history of the eastern Eurasian blocks (e.g. Lin, Fuller & Zhang 1985; Opdyke *et al.* 1986; Zhao & Coe 1989; Zhao *et al.* 1990a,b; Yang *et al.* 1991; Enkin *et al.* 1992). These studies have concluded that the NCB and SCB were welded by the Middle to Upper Jurassic. However, some dispute remains about the tectonic effects of the collision and penetration of India into Eurasia from the palaeomagnetic point of view (e.g. Lin *et al.* 1985; Li 1990; Lin & Fuller 1990; Zhao *et al.* 1990a,b). Lin & Fuller (1990) concluded that a large displacement of several thousands of kilometres had taken place between the SCB and northern Eurasia. On the other hand, it is also important to establish the sequence of events involved in the collision of the NCB and the SCB, in relation to the formation of the eastern portion of China since the Late Permian (e.g. Zhao & Coe 1989). Studies by Kent *et al.* (1986) and Zhao (1987) have identified probable widespread recent remagnetization in eastern China. The previous palaeomagnetic results from the eastern margin of China, i.e. the Shandong area of the NCB and the Zhejiang area of the SCB, therefore, need to be reviewed critically. It appears that the palaeomagnetic

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results from the western stable regions of both the NCB and SCB play an important role in elucidating post-collision tectonics. As part of a programme to develop an APW path for the NCB for Permian and younger times, palaeomagnetic sampling has been conducted in the Lower Cretaceous sequences located in the south–southwestern part of the Ordos basin, between Binxian (35.0°N, 108.0°E) and Jingchuan (35.3°N, 107.2°E) counties of Shaanxi Province (Fig. 1a). The results are compared with available data from both the North and South China blocks.

GEOLOGICAL SETTING AND PALAEOMAGNETIC RESULTS

The Ordos basin is the largest Mesozoic non-marine interior sedimentary basin in the NCB. Some 85 per cent of the basin has been covered by Quaternary loess. Geological investigations of this basin demonstrate that it has been stable with respect to western Pacific tectonic belts since the Permian (Ren *et al.* 1990). Lower Cretaceous continental sediments of the Zhidan Group are well developed in the southwestern part of the Ordos basin where they have a total thickness of up to 2000 m (Hao *et al.* 1986). According to extensive palaeontological research, the sediment series in the Binxian area consists of six formations. This series is composed mainly of red and purple to grey sandstone, mudstone interbedded with conglomerates, and purple mudstone and argillaceous sandstone in the lower part. The middle part includes interbedded grey mudstone, argillaceous sandstone and sandstone, grading upward to grey–yellow massive arkosic sandstone interbedded with siltstone. These sediments contain abundant ostracods, plant fossils and vertebrates such as *Lycoptera* sp., *Sinamia zdanskyi* Stensio, *Sinemys* sp., *Eotomistoma multidentato* Young, *Psittacosaurus* sp. and *Protoceratopus* sp., which are indicative of an Early Cretaceous age (Shaanxi Regional Stratigraphic Team 1983).

The Lower Cretaceous Zhidan Group was sampled in Binxian county, along the main road to Jingchuan county, in rising stratigraphic order. The samples were collected using a portable petrol-powered core drill, and oriented with a magnetic compass. At each site, 9 to 12 oriented cores were drilled over a stratigraphic thickness of 3–10 m. A total of 112 samples had been collected. The sites were collected along a 1500 m thick contiguous sequence. Bedding attitudes are gentle in this region.

Samples (2.2 cm in length) were measured and demagnetized at the palaeomagnetic laboratory of the Chinese Academy of Geological Sciences. Each sample was demagnetized by heating using a CAT-82 thermal demagnetizer built in cooperation with the laboratory of palaeomagnetism, Australia National University. In general, the specimens were heated to 300 °C in steps of 100 °C and then to 680 °C in steps of 50 °C, being measured with a Digico magnetometer after each step. The palaeomagnetic directions were analysed by principal component analysis (PCA; Kirschvink 1980) and the site mean direction was calculated using the statistical methods of Fisher (1953) or McFadden & McElhinny (1988). The computer program package of Enkin (1990) was used.

The intensity of natural remanent magnetization is of the order of about 10 mA m^{-1} for the red bed specimens. Most specimens showed a multicomponent magnetization (Fig. 2). A lower temperature component is, in most cases, close to the present Earth field (PEF) direction, and is completely removed by 300 °C (Fig. 2). A distinct high-temperature component was isolated between 350° and 680 °C, indicating that this component is probably carried by haematite. However, some of the grey sandstones showed a change of magnetic mineralogy during heating, and some showed changes of remanent vectors along great circles. The remagnetization circles with sector constraints were combined with directions obtained from PCA in order to calculate site mean directions (McFadden & McElhinny 1988) (Fig. 3a). The site mean characteristic remanent magnetization (ChRM) vectors have NNE downward, and antipodal SSW upward directions (Fig. 3b). The means of the normal and reversed directions are approximately anti-parallel (with an angular separation of 169.2° ± 14.3°) and thus the data appear to satisfy the positive reversal test (McFadden & McElhinny 1990).

The overall mean of 10 sites yields a direction of Dg = 18.6°, Ig = 51.5°, (α_{95} = 7.0°) in geographic coordinates, and Ds = 16.4°, Is = 50.5° (α_{95} = 6.8°) in stratigraphic coordinates (Table 1). The effect of the tilt correction is insignificant because of the gentle stratigraphic dip (less than 5°). Of the 67 samples exhibiting stable high-temperature components, however, 30 have reversed polarity showing at least 12 successive intervals of either normal or reversed polarity in this section that are roughly comparable with the Lower Cretaceous polarity sequence. We, therefore, believe that the ChRM directions probably are primary.

DISCUSSION

In recent years a considerable amount of new palaeomagnetic data have been obtained from China using modern techniques, covering the interval from Cambrian to Tertiary (e.g. McElhinny *et al.* 1981; Lin 1984; Kent *et al.* 1986; Opdyke *et al.* 1986; Zhao 1987; Li 1990; Zhao *et al.* 1990b; Enkin *et al.* 1991; Yang *et al.* 1991, 1992; Zheng *et al.* 1991). The results have been interpreted in terms of plate reconstructions and collision history. Widespread Late Mesozoic to Recent remagnetization related to the Yanshanian orogeny in the eastern part of China has been postulated (Courtillot & Besse 1986; Zhao, 1987; Kent *et al.* 1987). Courtillot & Besse (1986) also argued for the possibility of large displacements along the Tancheng–Lujiang fault. In either case, whether large-scale displacements or rotations of the NCB and SCB have occurred remains an open question, and has been discussed on the basis of sparse Late Mesozoic palaeomagnetic data coming mostly from eastern China (e.g. Lin *et al.* 1985; Lin & Fuller 1990; Zhao *et al.* 1990a). A selection and re-evaluation of the available palaeomagnetic results for the NCB and SCB since the Cretaceous has been carried out using the reliability criteria of Van der Voo (1990). The data passing these criteria are listed in Table 2 and the definition of the APW paths for the NCB and SCB is discussed below.

The APW path for the NCB since the Cretaceous

The overall mean ChRM direction defined in the present study corresponds to a Lower Cretaceous apparent pole

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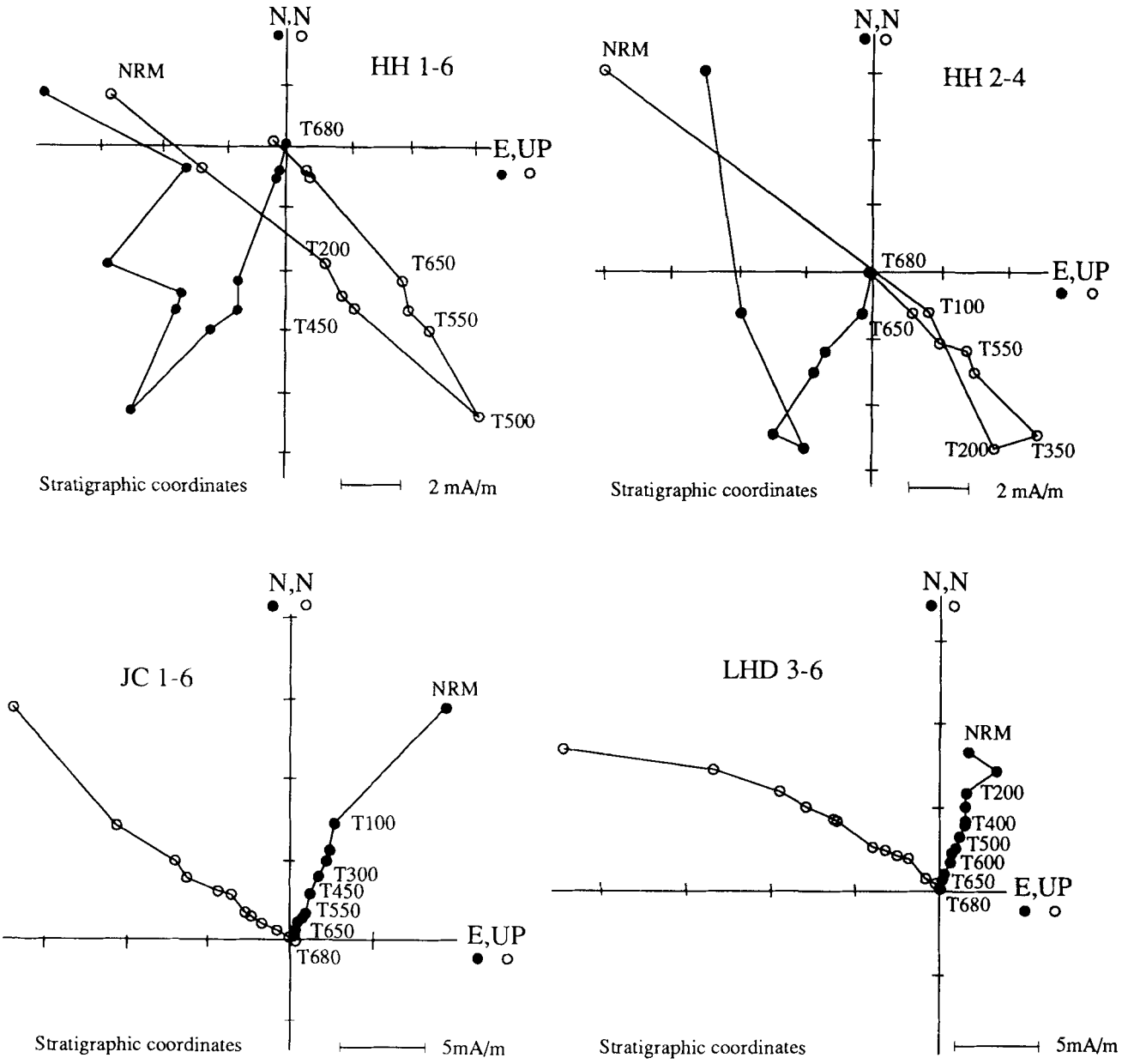


Figure 2. Orthogonal projections of thermal demagnetization data for representative samples, shown in stratigraphic coordinates.

position at 75.8°N, 20.7°E for the NCB (Fig. 4). This pole is statistically indistinguishable from the results for the northern part of the NCB situated in Central Asia and the Mongolian fold belts (Zhao *et al.* 1990b; Pruner 1992). However, the latter two results cannot be considered as reference data for the Cretaceous APW path of the NCB, for the simple reason that the site localities are mostly outside of the NCB. Furthermore, these two high-latitude Cretaceous poles are suspect because of possible incomplete removal of a secondary overprint and uncertain age of volcanic units. Pruner (1992) determined three site mean ChRM directions from two sites of volcanic lava and one sandstone site in the southern part of Mongolia. Given the secondary nature of the magnetization at the normal polarity sites, only the reversed polarity was interpreted as primary in origin. Similar pole positions for their basalt sites and the Neogene basalt results from the NCB (Zheng *et al.* 1991) (Table 2) suggest that the rock units may be Tertiary in age. The pole position calculated from the sandstone site of Pruner (1992) with $n = 30$ specimens is at 72.5°N and 205.3°N ($d_p = 6.4°$, $d_m = 8.6°$) which is very close to our Lower Cretaceous pole for the Ordos basin, and a Lower Cretaceous composite from both the NCB and Mongolia, as reported by Zhao *et al.* (1990b). The APW path for the NCB is, therefore, obtained by combining the result from the present study with the more reliable Upper Cretaceous and Neogene poles from Shanxi Province (Zheng *et al.* 1991) (Fig. 4a).

Palaeomagnetic poles obtained from post-Cretaceous rocks from the region located east of the Tancheng–Lujiang

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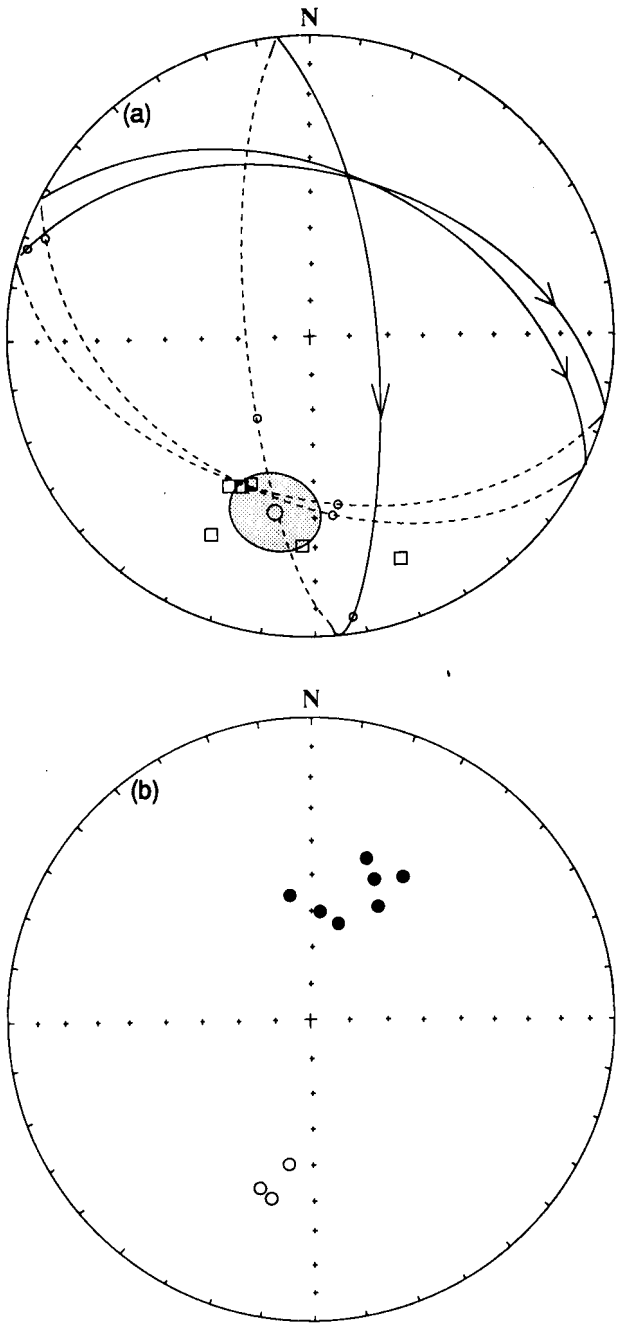


Figure 3. (a) An example of combined direct observation directions (open squares) and great circles with sector constraint directions (small open circles) for site HH2. The mean direction is also shown with its 95 per cent confidence (darker shading). (b) Equal-area stereographic projection of the Early Cretaceous site mean directions found in this study, after tilt correction, showing positive reversal test. Open and solid circles correspond, respectively, to the upper and lower hemisphere projections.

fault (Lin 1984; Lee *et al.* 1987; Uchimura 1989) are plotted in Fig. 4(a) (squares) for further comparison. These poles lie systematically to the east of the APW path for the western stable part and may indicate a possible clockwise rotation of the area located to the east of the Tancheng–Lujiang fault since the Tertiary. In the northern part of the Banxi areas (Uchimura 1989), the amount of rotation may have been as great as $29° \pm 10°$ since the Late Cretaceous. Such a clockwise rotation has also be recognized by Yang *et al.* (1991, fig. 12b) on the basis of Triassic results. The same phenomenon was also found in the Korean peninsula, where Lee *et al.* (1987) suggested a clockwise rotation of $9° \pm 11°$ relative to the Chinese blocks in post-Cretaceous time.

Table 1. Characteristic site mean directions for the Lower Cretaceous of the Binxian section.

Site	N(n,c)	N/N'	R/N	Dg	Ig	Ds	Is	ks	α_{95}
JC1	10(10,0)	10/10	0/10	37.4	57.0	30.8	54.8	35.1	3.8
LHD4	5(5,0)	5/8	0/5	35.9	45.5	32.4	43.8	48.6	11.1
LHD3	6(6,0)	6/8	5/1	192.6	-52.7	189.3	-49.7	9.0	23.9
LHD2	8(7,1)	8/9	2/6	11.7	61.6	4.6	60.8	25.5	11.2
HH3	3(3,0)	3/7	0/3	22.8	61.7	15.8	63.4	54.4	16.9
HH2	9(6,3)	9/10	9/0	191.3	-38.5	192.9	-38.8	21.7	11.5
HH1	10(9,1)	10/10	10/0	197.3	-41.1	197.3	-41.1	36.4	8.2
LH3	3(3,0)	3/3	0/3	23.7	47.9	23.7	47.9	42.3	19.2
LH2	7(4,3)	7/8	3/4	20.0	47.9	19.1	43.0	11.9	18.9
LH1	6(6,0)	6/8	1/5	351.6	55.1	350.7	55.9	127.9	5.9
Mean of Rev. sites			3/0	193.8	-44.1	193.4	-43.2	159.4	9.8
Mean of Nor. sites			0/7	21.2	54.7	18.0	53.6	46.4	9.0
Average			10	18.6	51.5	16.4	50.5	50.8	6.8

N, number of specimens (n, number of direct observations; c, number of remagnetization great circles); N/N', number of specimens used/number specimens measured; R/N, numbers of reversed/normal polarity specimens; Dg, Ig and Ds, Is, declination and inclination in geographic and stratigraphic coordinates respectively; ks, precision parameter after tilt correction; α_{95}, half-angle of cone of 95 per cent confidence about the mean direction in stratigraphic coordinates.

The Cretaceous pole from the Ordos basin of the NCB also differs significantly from the published Cretaceous data from the Siberian plate (Khramov *et al.* 1985). This may imply that an important tectonic event, such as intracontinental deformation (involving some 1200 km of shortening) has occurred in post-Cretaceous time. There is no clear geological evidence for such a phenomenon (Lee *et al.* 1987; P. Tapponnier, private communication, 1990). Alternately, Courtillot & Besse (1986) and Besse & Courtillot (1991) suggested that palaeomagnetic data from Siberia might be unreliable. A synthetic Eurasian APW path for the period since 200 Ma in 10 Ma intervals, using the transfer of the most reliable poles from Eurasia, North America, Africa and India according to ocean kinematics, was proposed by these authors. A modified version which uses no Asian data in order to avoid circular reasoning has been used here (J. Besse, private communication, 1990). An important question is whether European data can be used for Siberia. Yang (1992) have suggested that post-Palaeozoic motions throughout the Urals and the western Siberian basin were negligible, based on lower Triassic poles from Siberia and Europe/North America (Van der Voo 1990). An important hairpin loop occurs during the Cretaceous (130–80 Ma) in the European APW path (Fig. 4b). Our Lower Cretaceous pole for stable NCB matches well and further confirms this hairpin loop (see also Enkin 1990). A good general coherency between the APW path for the NCB and Eurasia since the Cretaceous is shown in Fig. 4(b). This indicates that the NCB has been welded to the Eurasian continent since at least the Early Cretaceous, and then maintained a stable position with respect to this continent. This conclusion contradicts that of Lin & Fuller (1990) and Li (1990) that the NCB/or Tarim block moved eastward some thousands of kilometres relative to Siberia in Tertiary times. Moreover, this would raise insuperable problems in

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Table 2. Palaeomagnetic pole data for the North and South China blocks since the Lower Cretaceous.

Nr.	Rock Type	Location (°N/°E)	Age	N	Pole position Lat(°N) Long(°E)	α_{95}	Reliability 1	2	3	4	5	6	7	Q	References
	North China Block														
	Red bed	35.0/108.0	K1	10	75.8/208.7	7.5	x	x	x		x	x	x	6	this study
	Red bed	37.4/114.5	K1	3	80.2/278.0	13.3	x	x			x	x		4	Xiong et al. 1987
	Red bed	36.6/109.0	K1	2	84.6/310.2	4.5	x	x			x			3	Cheng et al. 1988
	Red bed	35.8/117.5	K1	/	78.4/194.8	10.0	x	x					x	3	Zhang et al. 1990
	Mean all Q			4	82.4/231.8	10.0									
1	Mean Q>4			1	75.8/208.7	7.5									
2	Limestone	40.1/112.9	K2	4	79.6/170.1	5.8	x	x	x		x	x	x	6	Zheng et al. 1991
	Basalts	41.1/114.6	N	49	83.4/119.2	6.8	x	x	x	x	x	x	x	7	Zheng et al. 1991
	Basalts	40.9/113.2	N	13	88.6/29.2	8.2	x	x	x		x	x	x	6	Zhao et al. 1990, quoted in Zheng et al. 1991
3	Mean Q>4		N	2	85.2/192.5	5.1									Zheng et al. 1991
	Eastern Tangcheng-Lujiang Fault Terrane														
4	Red bed	35.9/119.4	K1	5	69.0/200.9	14	x	x	x			x	x	5	Lin 1984
5	Red bed	35.9/128.5	K1	19	67.6/205.1	5.8	x	x	x	x		x	x	6	Lee et al. 1987
6	Sandstone	41.3/123.8	K2	5	59.3/202.6	6.0	x	x	x	x			x	5	Uchimura 1989
7	Red bed	39.0/125.8	K2	2	69.0/178.0	6.0	x		x				x	3	Gurarri et al. 1966
8	Basalts	36.6/118.5	N	3	76.4/178.0	8.0	x	x	x			x	x	5	Lin 1984
	Central Asia-Mongolia Fold Belt														
	Volcanics	42.0/119.2	K1	6	82.9/249.5	5.7	x	x	x		x	x	x	6	Zhao et al. 1990
	Basalts, sandstone	45.4/107.6	K1	3	86.9/252.8	14.2		x	x		x	x		4	Pruner 1992
	South China Block														
	Red bed	29.7/120.3	K1	7	77.1/227.6	5.5	x	x	x		x	x	x	6	Lin 1984
	Red bed	30.0/103.0	K1	23	74.5/229.0	3.2	x	x	x	x		x	x	6	Enkin et al. 1991
	Red bed	30.0/103.0	K1	8	78.6/201.0	15.3	x	x	x			x	x	5	Zhuang et al. 1988, superceded by Enkin 1991
	Rhyolites	18.4/109.0	K1	/	80.0/246.4	/	x						x	2	Yu et al. 1990
	Red bed	30.1/103.0	K1	9	76.3/265.5	11.1	x	x	x	x		x	x	6	Otofuji et al. 1990, excluded for all Q mean
	Red bed	27.9/102.3	K1	2	77.4/196.2	15.0	x		x				x	3	Zhu et al. 1988
	Red bed	26.0/117.2	K1	/	66.9/221.4	5.4	x						x	2	Zhai et al. 1991
	Mean all Q			6	76.2/220.2	5.2									
9	Mean Q>4			3	77.0/220.6	6.1									
	Red bed	25.0/116.4	K2	/	67.9/186.2	9.2	x	x	x	x	x	x	x	7	Hu et al. 1990, magnetostratigraphic study
	Red bed	32.0/119.0	K2	10	76.3/172.6	10.3	x	x	x	x	x		x	6	Kent et al. 1986
	Granites	22.0/114.0	K2	12	78.2/171.9	10.6		x	x	x	x	x	x	6	Chan 1991
	Red bed	26.5/102.3	K2	10	80.9/296.8	7.7	x		x		x		x	4	Kent et al. 1986, excluded for all Q mean
	Red bed	27.0/115.0	K2	/	79.2/158.6	1.0	x	x			x	x	x	5	Gu and Gu 1990, magnetostratigraphic study
	Red bed	30.0/102.9	K2	16	72.8/241.1	5.0	x	x	x			x	x	5	Enkin et al. 1991, excluded for all Q mean
	Red bed	23.0/115.0	K2	6	66.0/221.5	3.0	x					x	x	3	Hsu 1987
	Red bed	27.7/120.9	K2	3	78.9/186.6	5.5	x		x		x		x	4	Zhu et al. 1988
	Red bed	19.6/109.5	K2	/	80.6/175.1	/	x						x	2	Yu et al. 1990
	Mean all Q			7	76.2/186.6	6.8									
10	Mean Q>4			4	75.6/175.0	6.7									
	Basalts	29.5/120.9	<10Ma	3	73.9/36.1	12.4	x					x	x	3	Lin 1984
	Red bed	30.1/102.9	E	6	85.2/173.8	14	x	x	x			x	x	5	Zhuang et al. 1988
	Red bed	24.7/102.9	E	/	83.8/309.1	10	x		x		x	x	x	5	Liang et al. 1986, magnetostratigraphic study
	Red bed	24.7/103.2	E	/	84.4/343.1	10	x		x		x	x	x	5	Liang et al. 1986, magnetostratigraphic study
	Mean all Q			4	85.5/11.3	10.9									
11	Mean Q>4			3	87.5/307.2	9.3									

N is the number of sites (or studies); ages of the rock units are indicated: K1, Lower Cretaceous; K2, Upper Cretaceous; E, Lower Tertiary; N, Upper Tertiary; Reliability criteria 1–7 are from Van der Voo (1990) as below: 1, well-dated rocks; 2, sufficient numbers of samples (>25) and high enough precision ($k > 10$ and $\alpha_{95} < 16°$); 3, detail of demagnetization and isolation of magnetic components; 4, positive field test; 5, sufficient structural control, i.e. no suspicion exists about possible local rotation; 6, presence of reversals; 7, lack of similarity with younger palaeopoles; Nr, numbers denoting palaeomagnetic poles in Fig. 4.

interpreting the NCB/SCB collision history recorded in the Qinling.

Comparison of the NCB, SCB and Eurasian APW paths since the Cretaceous

The available Late Mesozoic data from extensive areas of the SCB have been studied recently (e.g. Lin 1984; Kent *et al.* 1986; Zhu, Hao & Zhao 1988; Gu & Gu 1990; Gu 1990; Hu, Li & Ma 1990; Chan 1991; Enkin *et al.* 1991). The mean obtained from data of high quality ($Q > 4$) is statistically indistinguishable from the coeval poles for the APW path of the NCB and the Eurasian synthetic APW path for the Lower Cretaceous (Table 2; Fig. 4c). Outlying poles indicative of local rotations (Kent *et al.* 1986; Otofuji *et al.* 1990) were excluded from the APW path compilation

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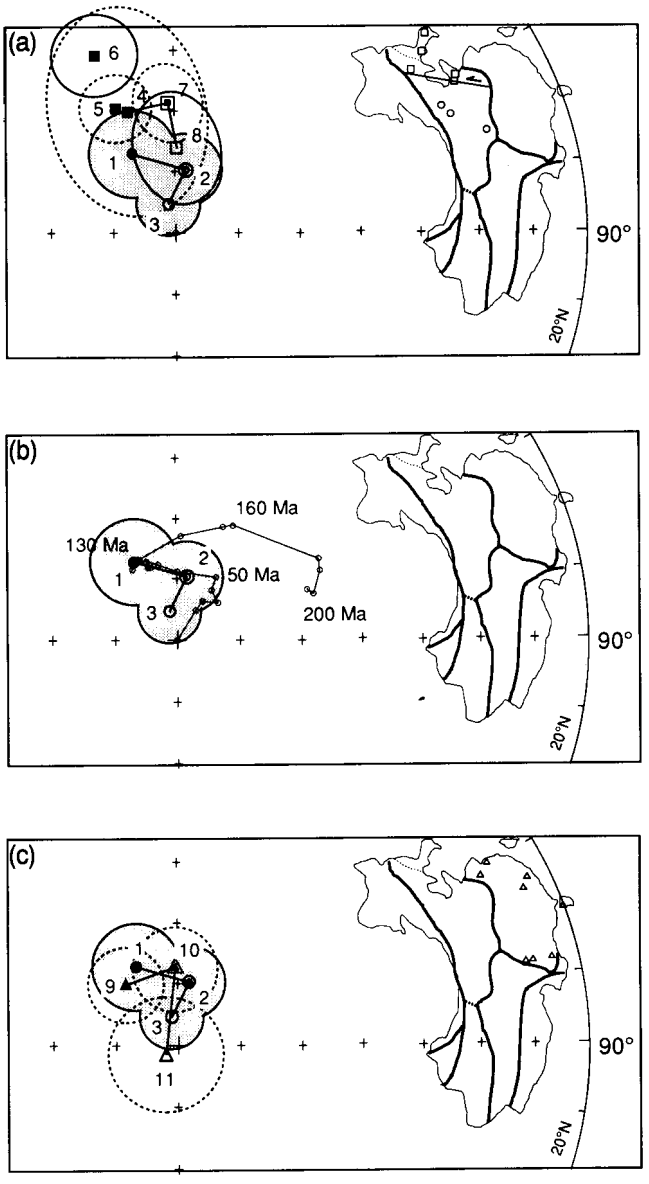


Figure 4. Comparison of the APW path for the NCB (circles) since the Late Mesozoic with (a) that of the eastern Tancheng–Lujiang fault area (squares), (b) the synthetic APW path for the Eurasian plate from 200 Ma to the present in 10 Ma intervals (Besse & Courtillot 1991) (open circles), and (c) APW path of the SCB (solid triangles). The corresponding sampled locations are shown on the simplified map. Numbers denoting the palaeomagnetic poles are consistent with Table 2.

for the south-west of the Sichuan basin. The Upper Cretaceous pole reported by Enkin *et al.* (1991) is located on the far side of the high-quality ($Q > 4$) mean pole (Table 2), which implies either the occurrence of northward intracontinental shortening (about 800 km) during the Tertiary in the Qinling fold belt or an error in the age of the rocks as suggested by the original authors (Enkin *et al.* 1991). However, magnetic inclination shallowing due to the coarse grain size and compaction effects cannot be ruled out (e.g. Butler, Dickinson & Gehrels 1991). The mean pole for the Upper Cretaceous of the SCB is also statistically indistinguishable from that of the NCB and the Eurasian synthetic pole for 50–70 Myr. The Palaeogene palaeomagnetic pole for the SCB is tentatively calculated from the results of a magnetostratigraphic study from the Yunnan Province (Liang *et al.* 1986) and Tertiary results from the western Sichuan basin (Zhuang *et al.* 1988) and is plotted in Fig. 4(c).

The similarity and almost complete overlap of the APW paths for the SCB, NCB and Eurasia is clearly evident in Figs 4(b) and (c). It implies that no resolvable relative movement has taken place between these blocks (within the uncertainties of the method). Palaeomagnetic applications to small-scale tectonic deformations are not a simple matter. However, further palaeomagnetic studies on a well-dated rock units from the stable regions of the south-east Asian blocks may possibly supply additional constraints needed for the interpretation of the tectonic evolution of this region (Fig. 1b).

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